

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.2: Chemicals of Life — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Driving Question	DRIVING QUESTION: What chemicals are in the food we eat, and what happens to them in our bodies?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Blood Glucose Puzzle: Why Does Glucose Rise and Fall After Eating Ugali?	We observed that Wanjiru's blood glucose concentration was 4.5 mmol/L before the meal of ugali, sukuma wiki and beans, rose to a peak of 8.0 mmol/L at 60 minutes (a rise of 3.5 mmol/L, about 78% above the start), then fell back to 4.5 mmol/L by 180 minutes (a fall of 3.5 mmol/L over 120 minutes). The rise was faster than the fall, and the level was back in the normal range (4.0 to 5.5 mmol/L) by 150 minutes at 4.7 mmol/L.	We learned how to read a blood glucose graph and table, to calculate the change in concentration and compare how fast it rose and fell, and to turn what we wonder into testable questions. We learned that the meal contains different chemicals (about 70 g carbohydrate in ugali, about 20 g protein and 15 g carbohydrate in beans, about 10 g fat from oil, and vitamins and mineral salts in sukuma wiki) and that we do not yet know which ones affect blood glucose.	We can explain, as a first idea, that something in the meal (probably the food chemicals in ugali) causes glucose in the blood to rise, and that something in the body must later lower it again because the level does not stay at 8.0 mmol/L. We cannot yet explain how food becomes glucose or where glucose goes, so we recorded these as questions on the Driving Question Board and in our initial flow-diagram model.
Lesson 2 Elements and Compounds of Life: What Is Wanjiru's Lunch Made Of?	What did we observe? Record the amounts of each nutrient in Wanjiru's meal (ugali about 70 g carbohydrate; beans about 20 g protein and 15 g carbohydrate; sukuma wiki about 3 g protein plus fibre, vitamins A and C, iron and calcium; about 10 g fat from cooking oil) and what happened when dry maize flour was burnt (it blackened, showing carbon).	What did we learn? Foods and living things are made of elements (C, H, O, N, P, S, Ca, Fe, Mg, Na, K) that form compounds. Organic compounds (carbohydrates, proteins, lipids, vitamins) contain carbon bonded to hydrogen; inorganic compounds (water, mineral salts) do not.	How can we explain Wanjiru's glucose graph so far? Her lunch contains several organic and inorganic chemicals, and the largest amount is carbohydrate (about 70 g in ugali), so carbohydrate is the most likely cause of the rise from 4.5 to 8.0 mmol/L. We still need tests and more evidence to show how it becomes glucose and why the level falls back.
Lesson 3 Carbohydrates: Starch and Sugars	Iodine turned blue-black with ugali and maize flour (starch present) but not with milk or nyama choma, and only slightly or not with sukuma wiki. Benedict's solution turned brick-red/orange with ripe banana	Carbohydrates are organic compounds. Starch is a large carbohydrate made of many glucose units; glucose is a reducing sugar. Ugali has about 70 g of carbohydrate, mostly starch from maize.	The starch in ugali is a source of the glucose that raised Wanjiru's blood glucose concentration from 4.5 to 8.0 mmol/L (a 3.5 mmol/L rise in 60 minutes), because starch is made of glucose units. We still need to

	and brick-red with glucose, green or yellow with milk, and stayed blue with ugali.	Ugali tested positive for starch but not for reducing sugar before digestion.	find out how starch is broken down into glucose and what happens to glucose after it reaches the blood.
Lesson 4 Proteins and Lipids: Do They Cause the Glucose Spike?	We observed that beans, milk and nyama choma turned violet/purple with the biuret test (protein), while ugali stayed pale blue. Groundnuts and mandazi left a translucent grease spot, groundnuts gave a cloudy white emulsion, and milk gave a faint result, while ugali was negative for lipids.	We learned that proteins are made of amino acids and lipids are made of fatty acids and glycerol, whereas carbohydrates are made of glucose units. Proteins are used for growth and repair, lipids for energy storage, and carbohydrates are the main energy source.	We explain that Wanjiru's blood glucose spike from 4.5 to 8.0 mmol/L at 60 minutes is mainly caused by carbohydrates (about 70 g in ugali), not by the about 20 g protein or about 10 g fat in her meal. Proteins and lipids are still used by the body, but not mainly to raise blood glucose.
Lesson 5 Water, Mineral Salts and Enzymes in Digestion	We observed that with saliva at about 37 degrees C the iodine test changed from blue-black at 0 min, to brown-black at 5 min, to yellow-brown at 10 min, and Benedict's test at 10 min was brick-red, while the control with boiled saliva or water stayed blue-black.	We learned that saliva contains amylase, an enzyme that is a protein and a biological catalyst, which digests starch to maltose and then glucose; water and mineral salts are inorganic chemicals that support this, and enzymes work best at about 37 degrees C, are denatured by boiling and inactive at 0 degrees C.	We can explain that the starch in Wanjiru's ugali is digested by enzymes such as amylase into glucose, which helps explain why her blood glucose concentration rises from 4.5 to a peak of 8.0 mmol/L at 60 minutes; we still need to find out how glucose gets into the blood and why it falls back to 4.5 mmol/L by 180 minutes.
Lesson 6 Absorption, Respiration and Regulation: Explaining the Fall Back to 4.5 mmol/L	We observed that after Wanjiru's meal of ugali, sukuma wiki and beans, blood glucose concentration rose from 4.5 to a peak of 8.0 mmol/L at 60 min (a rise of 3.5 mmol/L) and then fell by 3.5 mmol/L over the next 120 min, to 6.5 at 90, 5.2 at 120, 4.7 at 150 and 4.5 mmol/L at 180 min. The rise is faster than the fall. In our role-play and data analysis we saw glucose enter the blood and then leave it.	We learned that starch in ugali is digested by enzymes such as amylase into glucose, which is absorbed into the blood. The pancreas releases insulin, which causes cells, the liver and muscles to take up glucose. Cells use glucose in respiration (glucose + oxygen gives carbon dioxide + water + energy, ATP released) and the liver and muscles store the excess as glycogen. Proteins and lipids are also digested and used for growth, repair and energy storage.	Blood glucose rises after the meal because starch and other carbohydrates in the food are digested into glucose and absorbed into the blood. It falls back to 4.5 mmol/L by 180 min because insulin from the pancreas causes glucose to be taken up by cells, used in respiration to release energy, and stored as glycogen in the liver and muscles. Carbohydrates are the main cause of the spike, while proteins and lipids from beans, milk, groundnuts and mandazi are used for growth, repair and energy storage. This answers our driving question about the chemicals in the food we eat and what happens to them in our bodies.

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